

DiaTool-DPO: Multi-Turn DPO for Tool-Augmented LLMs

Enhancing tool-use dialogue flow via automatic preference trajectory construction

DiaTool-DPO

Direct Preference Optimization for TA-LLM Dialogue Control

Target audience: ML researchers and NLP engineers (tool-augmented LLMs, RLHF alignment)

SFT Alone Fails to Teach Slot-Filling and Tool Rejection

Core issue: SFT on expert traces does not robustly teach dialogue control policies.

Failure mode 1: missing follow-up questions for incomplete arguments.

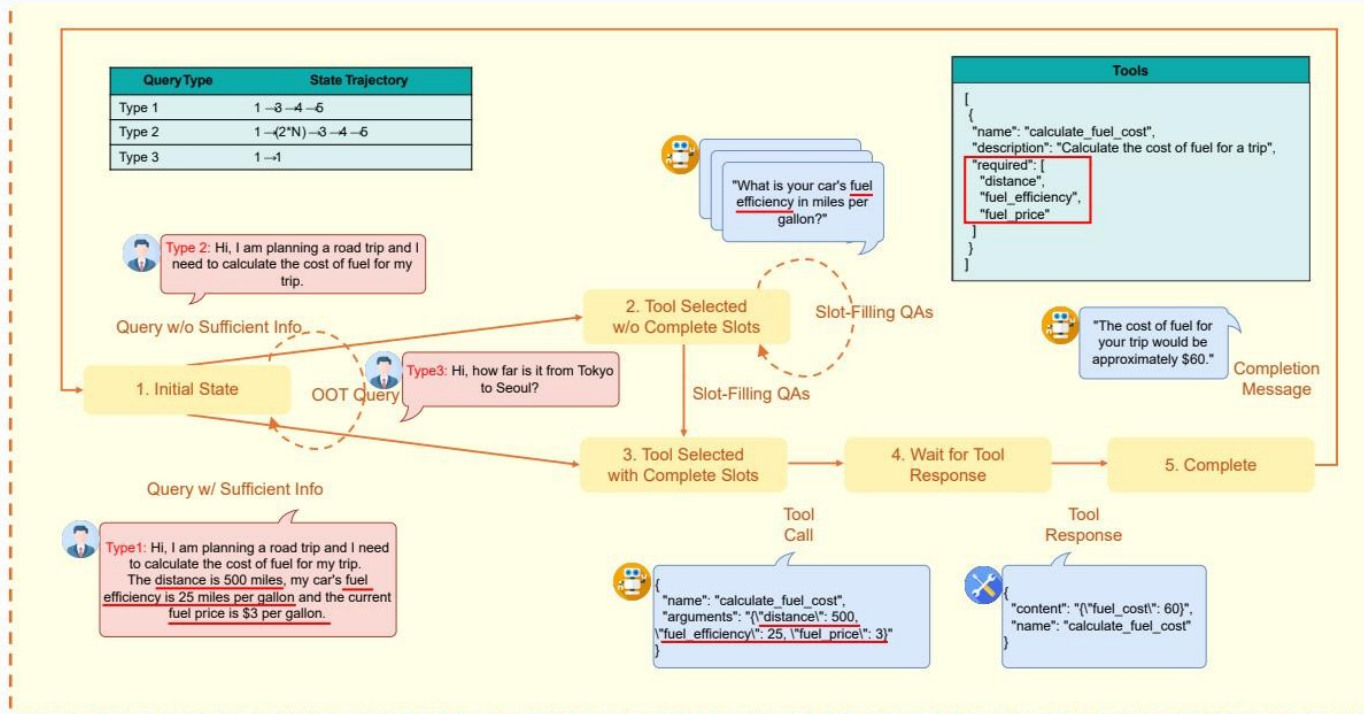
Failure mode 2: hallucinated tool invocations with fabricated slots.

Failure mode 3: failure to reject out-of-scope requests (unsafe tool calls).

Why preference optimization?

- Aligns model toward better dialogue trajectories, not just token mimicry.
 - Explicitly distinguishes correct vs. incorrect tool-use flows.
 - Enables policy shaping for ask/call/reject decisions across turns.
- DiaTool-DPO introduces automatic pair construction without human labels.

Modeling TA-LLM Dialogue as a 5-State Markov Decision Process



1 Initial: no history; reject out-of-scope requests.

2 Tool selected, slots incomplete; perform slot-filling QA.

3 Tool selected, slots complete; ready to invoke.

4 Waiting for tool response after execution.

5 Complete; generate final user-facing answer.

Three Query Types Define Distinct State Trajectories

Query Type	State Trajectory	Brief Example
Type 1	$1 \rightarrow 3 \rightarrow 4 \rightarrow 5$	User provides all required arguments, model calls tool immediately and returns result.
Type 2	$1 \rightarrow (2 * N) \rightarrow 3 \rightarrow 4 \rightarrow 5$	User query misses required slots; model asks follow-up questions N times before tool call.
Type 3	$1 \rightarrow 1$	Request is out-of-scope for available tools; model rejects tool call and stays in initial state.

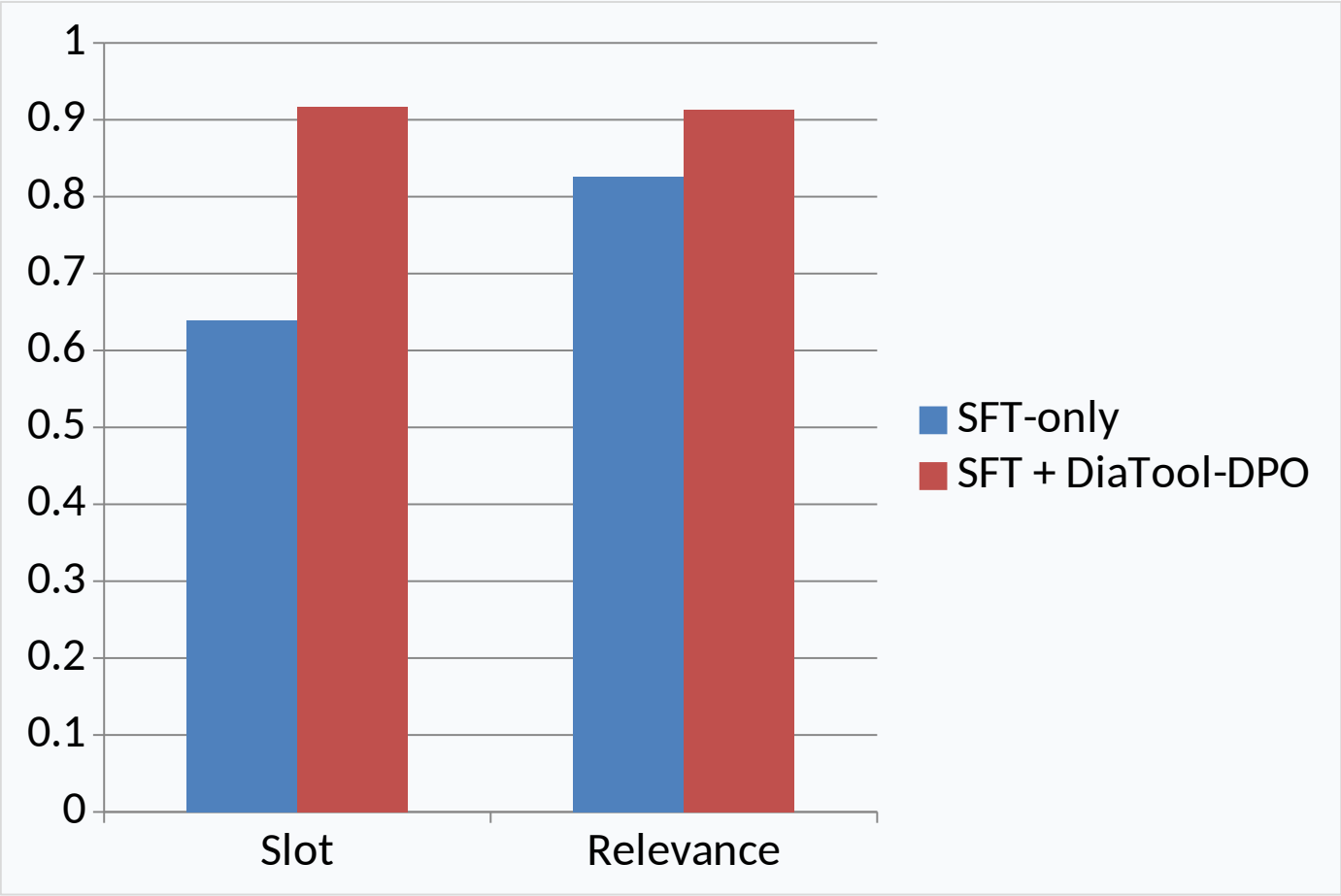
Automatic Paired Trajectory Construction Without Human Labels

Query	Chosen traj.	Rejected traj.	Learning lesson	Count	Difficulty
Type 1	1→3→4→5	1→2→3→4→5	Prevent redundant slot-filling	2,089	Easy
Type 1	1→3→4→5	1→(2*N)→3→4→5	Prevent redundant slot-filling	562	Hard
Type 1	1→3→4→5	1→(2*M)→3→4→5	Prevent redundant slot-filling	2,530	Hard
Type 1	1→3→4→5	1→1	Tool call accept	2,090/562	Easy, Hard
Type 2	1→2→3→4→5	1→3→4→5	Prevent slot hallucination	2,089	Easy
Type 2	1→(2*N)→3→4→5	1→3→4→5	Prevent slot hallucination	562	Hard
Type 2	1→(2*N)→3→4→5	1→(2*M)→3→4→5	Prevent slot hallucination	2,530	Hard
Type 2	—	1→1	Tool call accept	2,089/562	Easy, Hard
Type 3	1→1	1→3→4	Tool call reject	567	Hard
Type 3	1→1	1→(2*N)→3→4	Tool call reject	562	Hard

SFT + DiaTool-DPO Achieves Best Macro Average Across All Models

Model	Method	Call	Completion	Slot	Relevance	Micro Avg	Macro Avg
Prop.-8B	DiaTool-DPO-only	0.314	0.700	0.833	0.609	0.575	0.614
Prop.-8B	SFT-only	0.900	0.916	0.694	0.913	0.870	0.856
Prop.-8B	SFT + DiaTool-DPO	0.886	0.929	0.833	0.826	0.884	0.868
Prop.-3.1B	DiaTool-DPO-only	0.357	0.551	0.528	0.391	0.455	0.457
Prop.-3.1B	SFT-only	0.771	0.817	0.750	0.826	0.790	0.791
Prop.-3.1B	SFT + DiaTool-DPO	0.743	0.871	0.833	0.826	0.765	0.818
LLaMA-3-8B	DiaTool-DPO-only	0.029	0.449	0.056	0.261	0.205	0.199
LLaMA-3-8B	SFT-only	0.843	0.957	0.639	0.826	0.844	0.816
LLaMA-3-8B	SFT + DiaTool-DPO	0.857	0.929	0.917	0.913	0.905	0.904

LLaMA-3-8B: +43.5% Slot Accuracy and +10.5% Relevance with DiaTool-DPO



Before vs. After (LLaMA-3-8B)

Slot: 0.639 → 0.917 (+43.5%)

Relevance: 0.826 → 0.913 (+10.5%)

Largest gain appears in multi-turn information gathering (slot-filling).

Combined objective improves balance across call, relevance, and macro quality.

Key Takeaways: RL-Based Dialogue Control for Production TA-LLMs

- MDP formulation (5 states, 3 query types) enables systematic trajectory-level supervision.
- Automatic chosen/rejected pair construction removes human preference-labeling bottlenecks.
- DiaTool-DPO complements SFT: DPO-only underperforms, but SFT + DPO delivers best macro balance.
- Largest practical gains occur in slot-filling and tool-call rejection behaviors.
- Approach is language-agnostic and supports near GPT-4o-level conversational tool use on smaller models.